

EDUSAMBAM ASSESSMENT SERIES

Practice Test: Congruence, Similarity, Constructions & Loci

PRINTABLE PRACTICE WORKSHEET

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Subject	Mathematics (Maths)	Topic	Practice Test: Congruence, Similarity, Constructions & Loci	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20

Total	45	—	100
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Section A — Multiple Choice

30 marks

1. Two shapes that are exactly the same shape AND the same size are called:

Basic

- A. Similar
C. Regular
- B. Symmetrical
D. Congruent

Answer: ☐ A ☐ B ☐ C ☐ D

2. Which congruence test requires all three sides to be equal?

Basic

- A. SAS
C. ASA
- B. SSS
D. RHS

Answer: ☐ A ☐ B ☐ C ☐ D

3. For the SAS congruence test, the given angle must be:

Intermediate

- A. Included between the two known sides
C. A right angle
- B. Any angle in the triangle
D. The largest angle

Answer: ☐ A ☐ B ☐ C ☐ D

4. The RHS congruence test can only be used for:

Intermediate

- A. Isosceles triangles
C. Right-angled triangles
- B. Equilateral triangles
D. Any quadrilateral

Answer: ☐ A ☐ B ☐ C ☐ D

5. Two similar triangles have a linear scale factor of 3. Their area ratio is:

Advanced

- A. 9
C. 6
- B. 3
D. 27

Answer: ☐ A ☐ B ☐ C ☐ D

6. Two similar solids have a linear scale factor of 2. Their volume ratio is:

Advanced

- A. 2
C. 8
- B. 4
D. 6

Answer: ☐ A ☐ B ☐ C ☐ D

7. Congruent shapes are a special case of similar shapes where the scale factor is:

Intermediate

- A. 0
C. 2
- B. 1
D. Undefined

Answer: ☐ A ☐ B ☐ C ☐ D

8. A formal geometric construction may only use:

Basic

- A. A protractor and ruler
C. A calculator
- B. A protractor only
D. A ruler and a pair of compasses

Answer: ☐ A ☐ B ☐ C ☐ D

9. To construct a perpendicular bisector, the compass radius must be:

Intermediate

- A. Exactly the length of the segment
C. More than half the length of the segment
- B. Less than a quarter of the segment
D. It does not matter

Answer: ☐ A ☐ B ☐ C ☐ D

10. The locus of points a fixed distance from a single point is a:

Basic

- A. Circle
C. Triangle
- B. Straight line
D. Square

Answer: ☐ A ☐ B ☐ C ☐ D

11. The locus of points equidistant from two fixed points A and B is:

Intermediate

- A. A circle around A
C. An angle bisector
- B. The line segment AB itself
D. The perpendicular bisector of AB

Answer: ☐ A ☐ B ☐ C ☐ D

12. The locus of points equidistant from two straight lines meeting at a vertex is:

Intermediate

- A. The perpendicular bisector of one line
B. The angle bisector between them
C. A line parallel to both
D. A circle

Answer: ☐ A ☐ B ☐ C ☐ D

13. The locus of points a fixed distance from a straight line segment (including its ends) is:

Advanced

- A. A single circle
B. Two parallel lines with semicircular ends (a "stadium" shape)
C. A single straight line
D. A square

Answer: ☐ A ☐ B ☐ C ☐ D

14. Construction arcs on a finished diagram should be:

Basic

- A. Erased before submitting
B. Measured with a protractor
C. Drawn in pen only
D. Left visible, as evidence of an accurate method

Answer: ☐ A ☐ B ☐ C ☐ D

15. Bisecting a constructed 90° angle gives an angle of:

Intermediate

- A. 90°
B. 30°
C. 45°
D. 60°

Answer: ☐ A ☐ B ☐ C ☐ D

Section B — True / False

20 marks

16. "SSA" (two sides and a non-included angle) is a valid congruence test on its own.

Advanced

☐ True ☐ False

17. Similarity requires all corresponding sides to be in the same ratio, not just one pair.

Intermediate

☐ True ☐ False

18. A linear scale factor of 3 gives an area ratio of 3 (not squared).

Advanced

☐ True ☐ False

19. Two triangles with two pairs of equal angles (AA) are similar.

Intermediate

☐ True ☐ False

20. Congruent triangles always have the same area.

Basic

☐ True ☐ False

21. A protractor is required in a formal ruler-and-compass construction.

Basic

☐ True ☐ False

22. The angle bisector of an angle divides it exactly in half.

Basic

☐ True ☐ False

23. The plural of "locus" is "loci".

Basic

☐ True ☐ False

24. The locus of points equidistant from two points is a circle.

Intermediate

☐ True ☐ False

25. Erasing construction arcs before submitting a diagram is good practice.

Intermediate

☐ True ☐ False

Section C — Fill in the Blank

20 marks

26. Two shapes with the same shape but a different size are called _____.

Answer:

27. The congruence test using two sides and the included angle is called _____.

Answer:

28. If two similar shapes have a linear scale factor of k , their area ratio is k _____.

Answer:

29. If two similar solids have a linear scale factor of k , their volume ratio is k _____.

Answer:

30. The similarity test using only two pairs of equal angles is called _____.

Answer:

31. A geometric construction uses only a ruler and a pair of _____.

Answer:

32. The line that crosses a segment at 90° through its exact midpoint is called the _____ bisector.

Answer:

33. The set of all points satisfying a given rule is called a _____.

Answer:

34. The locus of points a fixed distance from a single point is a _____.

Answer:

35. Bisecting a constructed 60° angle gives an angle of _____ degrees.

Answer:

Section D — Column Matching

10 marks

36–40. Match each term to its correct definition.

Column A	Ans.	Column B
SSS	_____	A. A line crossing a segment at 90° through its exact midpoint
RHS	_____	B. All three sides of two triangles are equal
Angle bisector	_____	C. The complete set of points satisfying a given rule
Perpendicular bisector	_____	D. A right angle, the hypotenuse, and one other side are equal
Locus	_____	E. A line that divides an angle exactly in half

Section E — Short Answer

20 marks

41. Explain why "SSA" is not a reliable test for triangle congruence.

42. Describe how the area and volume of two similar shapes relate to their linear scale factor.

43. Describe the construction method for a perpendicular bisector of a line segment.

44. Explain what is meant by "the locus of points equidistant from two lines".

45. Describe the shape of the locus of points a fixed distance from a straight line segment.

Candidate Signature: _____

Invigilator Signature: _____

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Marks Obtained: _____ / 100 Checked By: _____

Result / Grade: _____ Date Checked: _____

Practice Test: Congruence, Similarity, Constructions & Loci — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. D | |
| 2. B | 9. C |
| 3. A | 10. A |
| 4. C | 11. D |
| 5. A | 12. B |
| 6. C | 13. B |
| 7. B | 14. D |
| 8. D | 15. C |

Section B — True / False

- | | |
|-----------|-----------|
| 16. False | 21. False |
| 17. True | 22. True |
| 18. False | 23. True |
| 19. True | 24. False |
| 20. True | 25. False |

Section C — Fill in the Blank

- | | |
|-------------|-------------------|
| 26. similar | 31. compasses |
| 27. SAS | 32. perpendicular |
| 28. 2 | 33. locus |
| 29. 3 | 34. circle |
| 30. AA | 35. 30 |

Section D — Column Matching

- | | |
|------------------------|--------------------------------|
| 36. SSS — B | |
| 37. RHS — D | 39. Perpendicular bisector — A |
| 38. Angle bisector — E | 40. Locus — C |

Section E — Short Answer (Model Answer Keywords)

41. two triangles; not always; ambiguous; two different
42. squared; cubed; area; volume
43. arc; compass; midpoint; equal radius
44. angle bisector; equal distance; equidistant
45. stadium; parallel; semicircle